Nihal Sirivolu

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Education

Virginia Polytechnic Institute and State University **May 2027**
B.S. in Computational Modeling & Data Analytics *Blacksburg, VA*

Experience

Edrevel AI **June 2026 – August 2026**
AI/ML Engineering Intern *Fairfax, VA – Hybrid*

- Engineered an **end-to-end computer vision pipeline** for workforce-training compliance using Python, PyTorch, OpenCV, YOLO, SAM-2, and MediaPipe to extract objects, people, hand interactions, and task evidence from video.
- Implemented **open-vocabulary detection, persistent tracking, segmentation, and keyframe extraction**, generating auditable bounding boxes, timestamps, scene assignments, and confidence metadata for reviewer workflows.
- Integrated ML inference with **FastAPI, Docker, and AWS**, connecting video uploads to backend processing, evaluator results, artifact storage, and review workflows.
- Automated **Jira REST workflows** for model-evaluation failures and test results, attaching evidence artifacts and preserving traceability from video scene to reviewer decision; validated changes against a **600+ test suite**.

IDPro AI Tourism – Virginia Tech **August 2026 – Present**
Research Team Member *Blacksburg, VA*

- Building an **Android AI-tourism app** with Kotlin and Jetpack Compose, combining CameraX capture, Android SpeechRecognizer, and spoken responses to help users explore their surroundings.
- Integrating **Ray-Ban Meta smart glasses** through Meta's Wearables Core and Camera SDKs, managing device registration, video streams, and photo capture, with testing through the Mock Device Kit.
- Instrumenting **Firebase Analytics** for interaction analysis and maintaining reproducible Gradle builds and dependency configurations for team development.

4SightAI **June 2025 – August 2025**
Automation/Cloud Engineering Intern *San Francisco, CA – On-Site*

- Built real-time ingestion services for health, exercise, dietary, and business data supporting AI products, reducing processing latency by 45% and improving measured model-context accuracy by 35%.
- Developed **Pandas and Power BI dashboards** for patient and operational datasets, translating data-quality and workflow signals into recommendations for product teams.
- Automated **Python and Jira workflows** for ingestion failures and data-quality checks, and implemented AWS Lambda, S3, and DynamoDB pipelines with vector retrieval for unstructured-data processing.

Projects

LNKZ MCP Server | *MCP, Node.js, Docker, AWS CDK* **August 2026 – Present**

- Building an **open-source Model Context Protocol server** for portable conversation context across LLM clients, exposing 24+ tools for import, search, context packets, and secure handoffs.
- Implemented **API-key authentication and scope-based authorization** with expiring, hashed, usage-limited, and revocable handoff tokens; verified workflows with automated tests and GitHub Actions.
- Authored and synthesized a **Docker-based AWS CDK deployment** using ECR, App Runner, and private RDS PostgreSQL with IAM roles, Secrets Manager, and KMS encryption; deployment remains staged for account rollout.

RSNA Knee Abnormality Detection | *Python, PyTorch, DICOM* **August 2026 – Present**

- Developed a **multimodal PyTorch pipeline for knee MRI abnormality classification**, integrating DICOM preprocessing, image augmentation, and multilingual radiology-report normalization.
- Investigated the contribution of **imaging and clinical-text features** through image-only, text-only, and multimodal baselines, using cross-validation, class-imbalance handling, and decision-threshold calibration.
- Established **configuration-driven training and inference workflows** to support reproducible experiments, comparative model evaluation, and controlled ablation studies.

Technical Skills & Certifications

Languages: Python, Kotlin, Java, C/C++, JavaScript, SQL
LLM / ML: Model Context Protocol (MCP), NLP, PyTorch, TensorFlow, OpenCV, YOLO, SAM-2, MediaPipe, Pandas, Power BI
Mobile / Wearables: Jetpack Compose, CameraX, Android SpeechRecognizer, Text-to-Speech, Meta MWDAT, Firebase Analytics
Backend / Cloud: Node.js, Express, FastAPI, PostgreSQL, Docker, GitHub Actions, AWS CDK, Lambda, S3, DynamoDB, Rekognition
Certifications: AWS Certified Solutions Architect – Associate; AWS Certified Cloud Practitioner